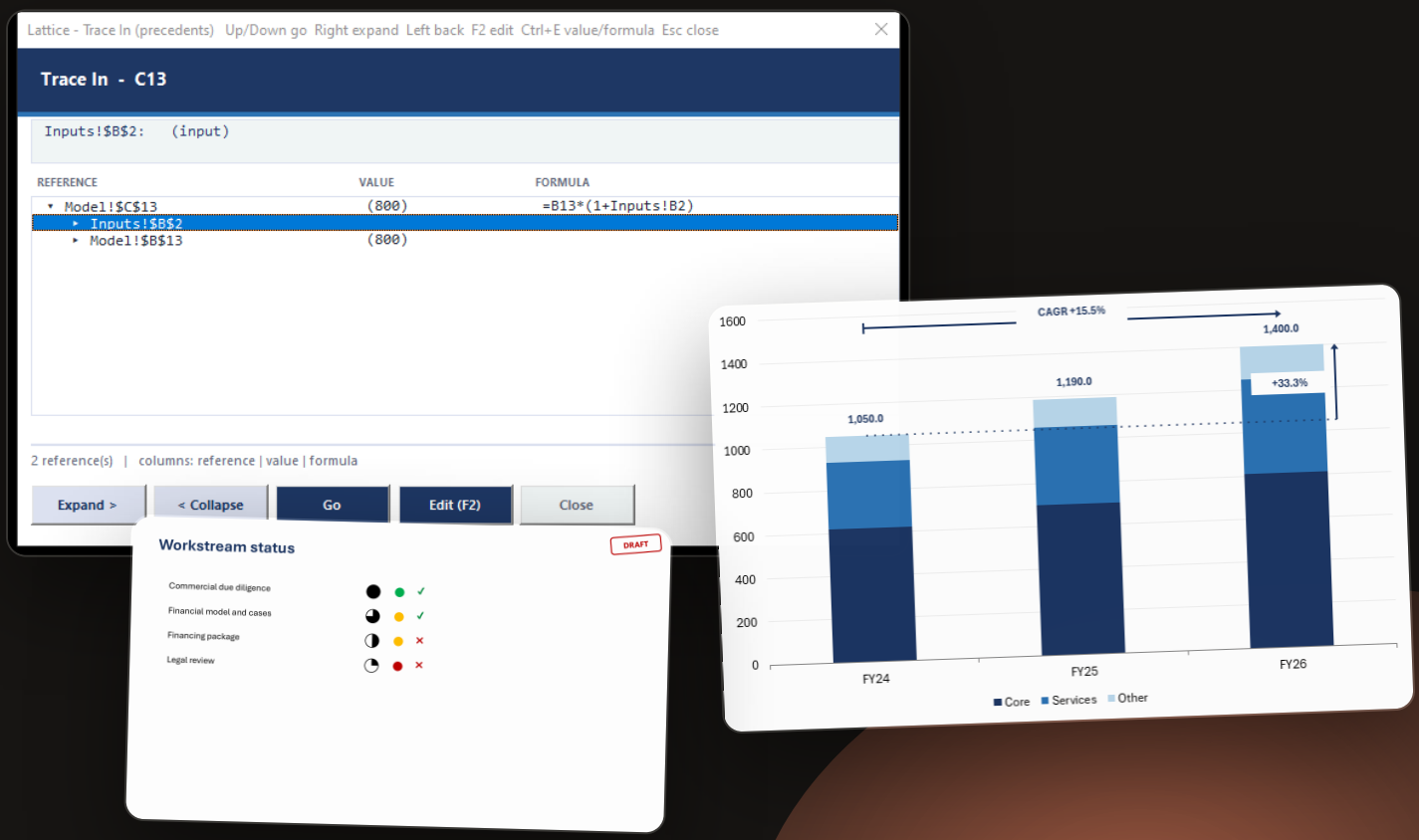
AN OPEN ADD-IN FOR EXCEL, POWERPOINT AND WORD

Lattice

The mechanics bankers, private-equity associates and hedge-fund analysts pay Macabacus and think-cell for, rebuilt in plain VBA with a PowerShell build. No licence server, no telemetry. Installs per user in a minute.

What it does | What it deliberately doesn't | Versus Macabacus and think-cell | How it was built



02 Why this exists 03 Audit without the mouse 04 Colour, cases, charts 05 Links that refresh 06 Slide hygiene
07 Versus Macabacus and think-cell 08 How it is built

Two subscriptions, one afternoon of mechanics

Every deal team runs on the same hundred gestures: colour the inputs blue, jump to the precedent, flip the sign, paste the table into the deck and refresh it before the meeting, align the boxes, stamp DRAFT on the title. Macabacus and think-cell sell those gestures. Lattice rebuilds them.

Nothing was decompiled or copied. The public feature lists and documentation screenshots of the commercial products were the specification; the implementation is original VBA, organised around the three things a working analyst actually touches all day: the model, the link into the deck, and the slide itself.

Lattice was written by one person directing an AI coding agent over a handful of sessions, each round installed and swept through COM-driven tests against real Excel, PowerPoint and Word before being used on live work. The repository history records what landed when, including the mistakes: a keytip Office silently refused, a hidden Excel instance that hijacked imports, a chart label that lost its name.

> *It is not that the mechanics are hard. It is that nobody had bothered to write them down in the open.*

THREE SURFACES, ONE DESIGN

One **Lattice** tab per app, opened with Alt K in Excel and Alt L in PowerPoint and Word. Everything Lattice draws follows one visual system: a navy and azure palette, tiled icons, windows with a title band and column headers, report sheets with an accent line and a frozen header, a Settings dialog per app.

402	37	3
RIBBON COMMANDS	EXCEL SHORTCUTS, REBINDABLE	ADD-INS, ONE TAB EACH

9.1k	25	~1 s
LINES OF VBA	COM TEST SCRIPTS	PER LINKED REFRESH

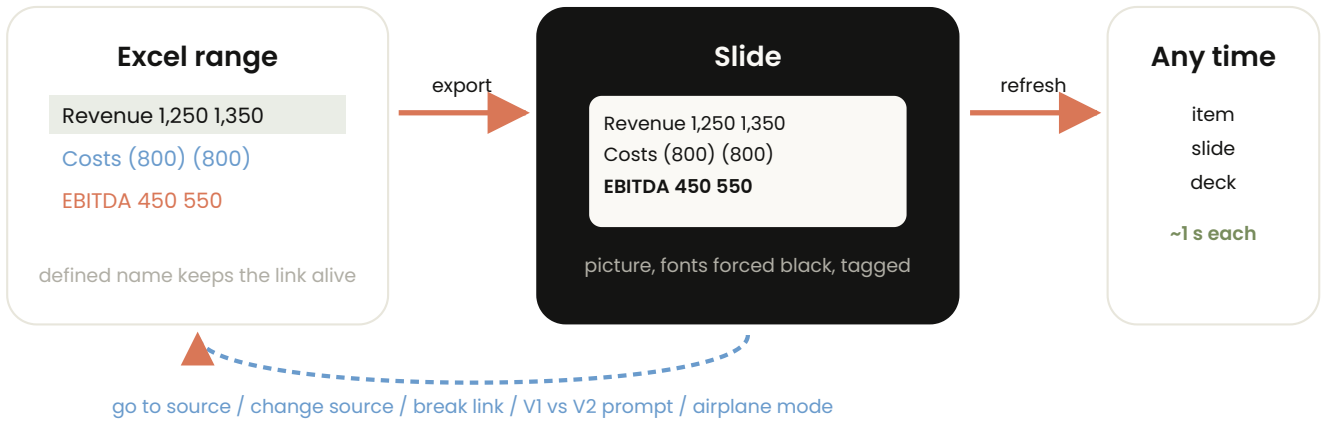
WHERE THE TIME GOES

- **Model**: colour, format, audit, cases, charts.
- **Link**: export once, refresh for the rest of the deal.
- **Slide**: align, conform, stamp, paginate, check.

HONEST STATUS

Personal-use quality on Microsoft 365 for Windows. Not every corner of the commercial tools is covered; page 7 lists the gaps.

HOW A LINK WORKS



HEDGE-FUND ANALYST

Earnings day: paste the KPI table into the note, refresh it after the model update, flip the sign on the bridge, walk the EPS build with Ctrl+[before the call.

PE ASSOCIATE

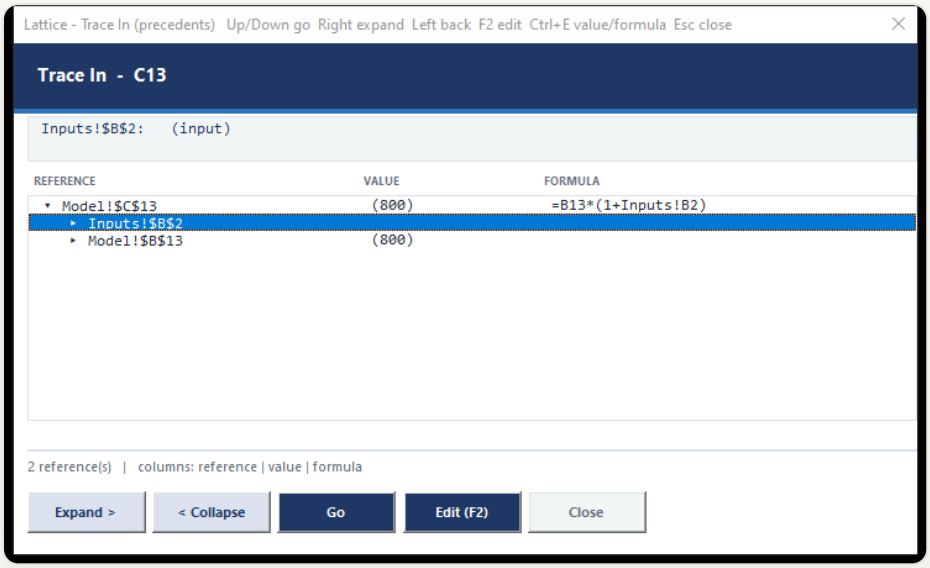
IC memo week: run Base, Upside and Downside into one table, football field for the valuation page, DRAFT on every slide until the partner signs off.

BANKER

Pitch at 2 a.m.: Auto Align the boxes, conform the title fonts, agenda and page numbers in one go, Deck Check before it goes to the printer.

Audit a model without touching the mouse

The jump keys do what Capital IQ's tracer and Macabacus's trace window do, and the window closes on Enter so the next key is F2.



Trace In. A tree of precedents with values and formulas. Up and Down move Excel live, Right expands a reference into its own inputs, Enter jumps to the cell and closes the window, Ctrl+E swaps the value and formula columns.

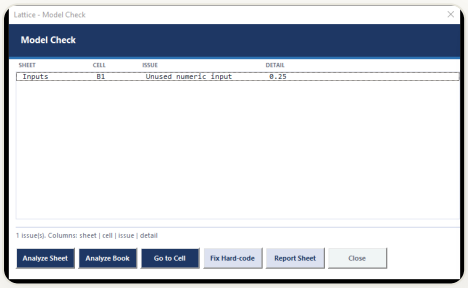
One press each

Ctrl [goes to the precedent of the active cell, Ctrl] to its dependent, Ctrl Shift Enter returns. When several candidates exist a picker opens; your arrow keys move through it while Excel follows.

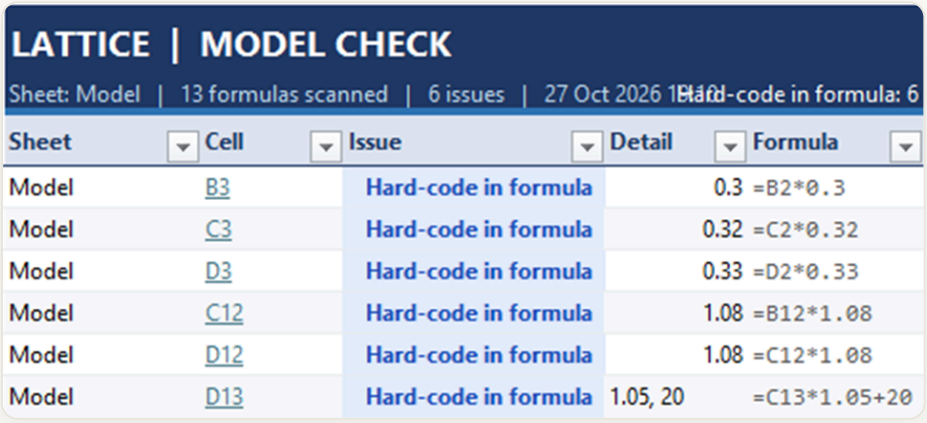
References are resolved explicitly through workbook and sheet, so cross-sheet jumps and sheet names with spaces work from any window.

Visualize

Highlight hard-codes Ctrl Shift K, formula flow Ctrl Shift L, select errors, inconsistent formulas and inputs, arrows on and off.



Model Check pane. Hard-codes inside formulas, inconsistent rows, errors and masked errors; one click jumps, Fix pulls the number into its own input cell.



Audit report. The same findings as a styled sheet with filters, counts and the offending formula, ready to send to the person who built the tab.

The audit loop, keyboard only

Ctrl [jump to the precedent

Ctrl] jump to the dependent

Ctrl Shift Enter back

Ctrl Shift [Trace In window: Up / Down follow, Right expands, Enter jumps and closes, F2 edits

MODEL CHECK FINDS

Hard-codes inside formulas (a 1.05 in a growth line), formulas that break the pattern of their row, #REF! and #DIV/0!, errors masked by IFERROR, numeric inputs nothing reads.

THE PICKER

When a cell has several precedents, a small window lists them with values and formulas. Arrow keys move Excel live; Enter or a double-click goes there and closes it.

TRACE OUT

The same tree in the other direction: who consumes this cell, down to the outputs linked into the deck. Ctrl Shift] opens it.

Colour, cases and charts that explain themselves

EUR m	FY24	FY25	FY26
Revenue	1,250	1,350	1,458
Costs	(800)	(800)	(820)
EBITDA	450	550	638

AutoColor. Blue inputs, black formulas, green links to other sheets, purple external links, dark red where a number is buried inside a formula (the *1.08* and the *+20* above). Selection, sheet or workbook, in bulk; a Black Fonts switch reverses it all for a print.

input formula sheet link external hard-code inside

Case	Base	Downside
case #3	Upside	
	Downside	

Case drop-down. The bare Forms combo every model hand-builds: input range = the case names, cell link = the selected number. Copies on other tabs keep working.

LATTICE | CASE RUNNER Case cell Drivers!B2 | 3 cases | 5 outputs

Output	Cell	Base	Upside	Downside	Upside vs Base	Downside vs Base
Revenue	Model!B17	1,250	1,400	1,050	+12.0%	-16.0%
EBITDA	Model!B18	350	434	231	+24.0%	-34.0%
Net debt / EBITDA	Model!B19	3.1x	2.6x	4.4x	-16.1%	+41.9%
IRR	Model!B20	18.0%	26.0%	7.0%	+44.4%	-61.1%
MOIC	Model!B21	2.3x	3.1x	1.4x	+34.8%	-39.1%

Case runner. Writes 1..n into the case cell, recalculates, reads the outputs you pick and tabulates them with change versus base, red and green, data bars.

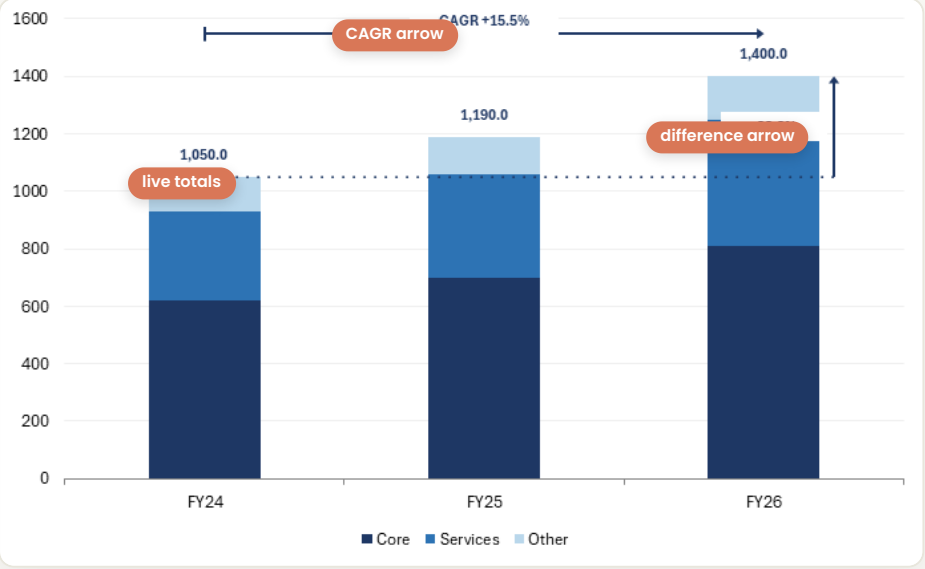


Chart annotations on native Excel charts. The totals are a real series driven by formulas on a hidden sheet, so they move with the data; the arrows are shapes inside the chart, so they follow it into the deck through the picture link.

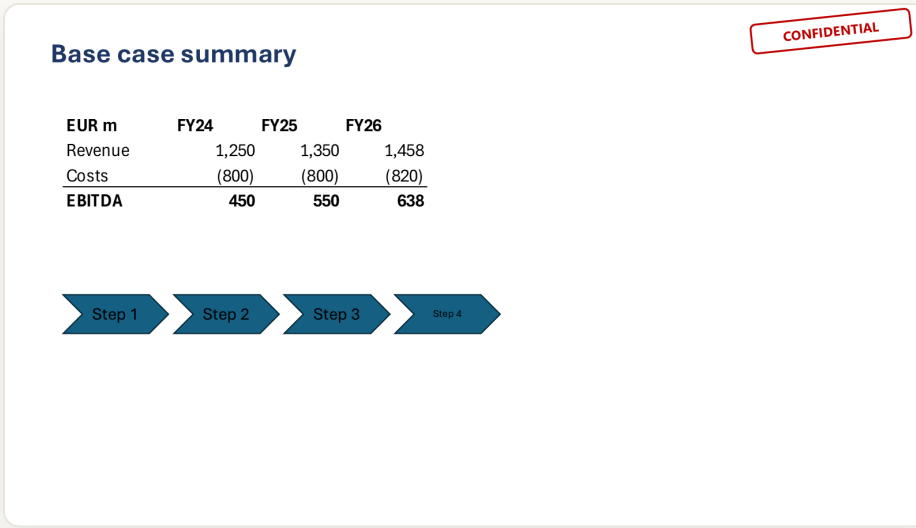
Formula keys

- Ctrl Shift Q cycle $ anchors
- Ctrl Shift N flip sign (toggles, no stacked minus)
- Ctrl Shift E IFERROR wrap and unwrap
- Ctrl Shift R / D fast fill to the header's edge
- Ctrl Shift M CAGR cell
- Ctrl Shift 1 ... 6 number, percent, multiple, date, currency cycles

LIBRARY

LBO · DCF · Distribution waterfall · Accretion / dilution · Returns IRR / MOIC · Debt schedule · Credit statistics · Black-Scholes · 2-way sensitivity · Scenario switch · Timeline · Run cases · Comps stats · Football field · Waterfall

Export once. Refresh for the rest of the deal.



A linked picture. The Excel block arrives with gridlines removed and every font forced black; the chevrons are Turbo Shapes; the stamp is a Sticker. Right-click the picture for Refresh this item, Go to source, Conform and Break link.

Five forms of export

- **Picture** Ctrl Alt R, fonts forced black, EMF or PNG
- **Editable table** Ctrl Alt T, keeps fills and borders
- **Chart** as vector picture Ctrl Alt C
- **Native PowerPoint chart**
- **Embedded workbook**

Match Width, Height or Size replaces the selected shape in the deck in place; new items land inside the import margin.

ALSO TO WORD

Picture or table at the cursor, Ctrl Alt W; refresh selection or document.

Links that survive

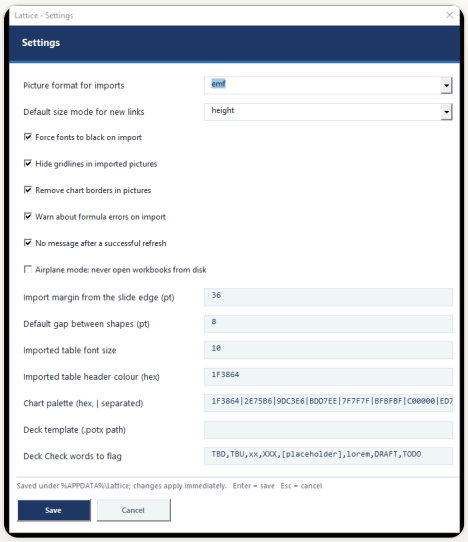
Each link carries the workbook, sheet, range and a defined name, so rows can move. When a V2 of the model is open instead of V1, one prompt repoints every link. When the file moved, a picker. In airplane mode nothing is opened from disk.

Fast by construction

The black-font pass runs inside Excel through Application.Run instead of cell by cell over COM: a 40×10 block refreshes in about a second, and the refresh binds to your visible Excel even when a hidden instance exists.

Text links

A number or a year inside a title can be tied to a cell: the sentence keeps its wording, the figure refreshes with the deck. Explorer pane lists every link; Highlight Links marks them in slide sorter.



Settings. Every option in one window, Enter saves.

Shapes

Align to the first shape or the slide, distribute, set gaps, same width / height / size, stretch, swap, nudge, arrange in a grid, **Auto Align** (snaps a loose selection to its implied rows and columns), align to the body area, arrange inside a container, stack, z-order, Select Match (same fill, font or size), convert to picture, crop to slide, remember and apply a position.

Keytips: Alt L then 1–6 for the six alignments, 7 8 distribute, 9 0 Q same width, height, size.

0.9 s
REFRESH OF A 40 X 10 BLOCK

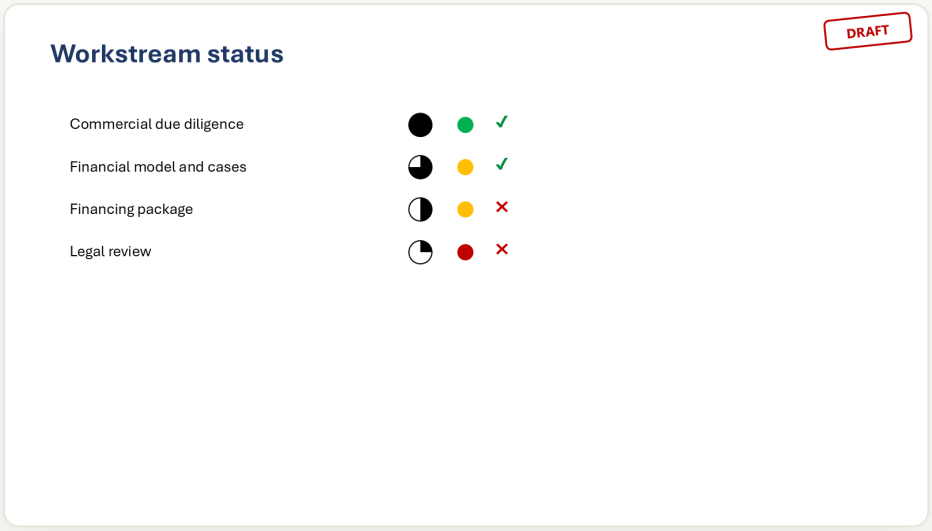
5
FORMS OF EXPORT

3
REFRESH SCOPES: ITEM, SLIDE, DECK

REFRESH, STEP BY STEP

Every link is read from its tags; the workbook is found among all running Excel instances before anything is opened from disk; the range is copied inside Excel with gridlines hidden and fonts forced black; the picture is pasted over the old one keeping position, size mode, z-order and name; the model's calculation mode and colours are restored. Formula errors in the range are reported once at the end.

Status marks, agendas and a deck that checks itself



Stickers. Harvey balls from Turbo Shapes cycle a quarter per click; traffic lights, ticks and crosses sit next to the selected shape; stamps (DRAFT, CONFIDENTIAL, PRELIMINARY, FOR DISCUSSION, ILLUSTRATIVE) go top-right, one per slide, replaced on re-run, DRAFT on every slide in one go, removed in one go.

Presentation

- **Agenda**: sections, divider slides and a table of contents kept in sync with the deck.
- **Pagination**: page numbers in your format, blank pages.
- **Deck Check**: leftover placeholders, forbidden words (TBD, lorem, XXX), off-slide shapes, mixed fonts.
- **Deck Tools**: harmonise and conform fonts across titles and subtitles, table formats, chart palette, a notes collector, per-section PDFs.
- Template, backup, PDF, email.

Turbo Shapes

Harvey balls, chevrons and chevron processes, timeline, numbered circles, sticky notes, callouts, rounded rectangles, arrows, brackets, braces, lines. Placed next to the selection, themed.

Styles and colour

Save the look of a shape and apply it to a selection or the whole deck; colour cycles, hex and theme colours, a format painter that survives across slides; text tools for case, bullets, superscripts and double spaces.

Library

A Slide Library with Slide Links: insert a slide from the library and later pull the library's version back, or push your edited slide into it. Master Shapes keeps your own shape library.

KEYTIPS IN POWERPOINT AND WORD

Alt L N new deck · Alt L G agenda · Alt L P pagination · Alt L K deck check · Alt L D deck tools · Alt L B library · Alt L J stickers · Alt L U turbo · Alt L M master · Alt L C colour · Alt L Y styles · Alt L S select match · Alt L W arrange · Alt L I import · Alt L T link text · Alt L R refresh · Alt L X explorer · Alt L H highlight · Alt L A airplane · Alt L O settings

DECK CHECK LOOKS FOR

empty placeholders · TBD / XXX / lorem · shapes off the slide · mixed fonts · missing page numbers · inconsistent titles

Run it before a deck leaves the building: every finding is listed with its slide number and selects the shape when clicked.

AGENDA IN PRACTICE

Sections become divider slides and a table of contents; rename or reorder a section and the contents page follows. Pagination writes "n / N" or your own format and skips the title and dividers if you ask. Deck Tools can print one PDF per section for the data room.

TURBO SHAPES AND STICKERS AT A GLANCE



chevron process · Harvey balls, cycle a quarter per click · lights, tick, cross · sticky note, stamp

Versus Macabacus and think-cell

Where the three overlap, Lattice is a peer. Where think-cell's chart engine starts, Lattice stops on purpose.

Capability	Macabacus	think-cell	Lattice
Formatting cycles and AutoColor	Yes	–	Yes
Precedent / dependent jumps, trace window	Yes	–	Yes
Hard-code and consistency audit	Yes	–	Yes
Model generators (LBO, DCF, returns, debt, sensitivity)	Partial	–	Yes
Case drop-down and case runner	No	–	Yes
Excel to PowerPoint / Word links with refresh	Yes	Partial	Yes
Match size / replace in place on export	Yes	–	Yes
Version-aware refresh (V1 vs V2 open)	Partial	–	Yes
Shape alignment, distribution, Auto Align	Yes	–	Yes
Agenda / table of contents, pagination	Yes	Yes	Yes
Slide library and slide links	Yes	–	Yes
Styles applied across a deck	Yes	–	Yes
Harvey balls, stickers, stamps	Partial	Yes	Yes
Chart totals, CAGR and difference arrows	Partial	Yes	Yes
Waterfall, football field, stacked with totals	Yes	Yes	Yes
In-slide chart engine with editable datasheet	–	Yes	No
Mekko and Gantt charts	No	Yes	No
Automatic label placement on charts	–	Yes	No
Workbook diff (two versions)	Yes	–	Partial
Settings dialog	Yes	Yes	Yes
Mac support	Yes	Yes	No
Licence server, telemetry, subscription	Yes	Yes	No

What Lattice deliberately does not do

- It does not replace PowerPoint's chart engine. Annotations go on native Excel charts and arrive in slides as linked pictures, or as native PowerPoint charts without annotations. think-cell's in-slide charts with automatic label layout, Mekko and Gantt are its real moat and are out of scope.
- No keyboard hook in PowerPoint (VBA cannot do it safely), so PowerPoint shortcuts are keytips.
- No cloud, no collaboration, no Mac.

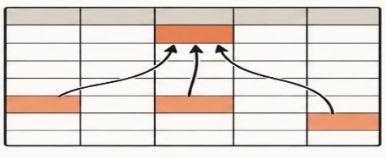
What it adds that neither has

- A case runner that turns the case drop-down into a scenario table in one command.
- A refresh that recognises a V2 of the model and repoints every link once.
- A trace window built for the keyboard: Enter closes it so F2 is the next key.
- Plain text settings and shortcuts you can read, diff and share.

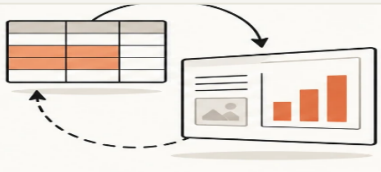
16
CAPABILITIES MATCHED OR EXCEEDED

1
PARTIAL: WORKBOOK DIFF

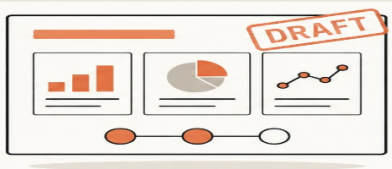
5
LEFT OUT ON PURPOSE



The model. Colour, audit, cases.

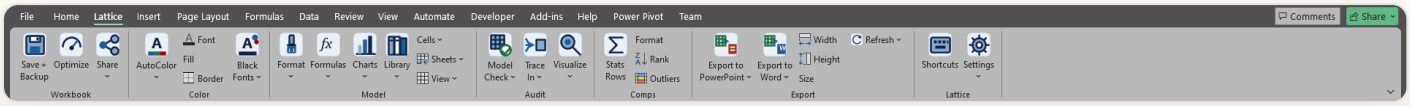


The link. Export once, refresh always.

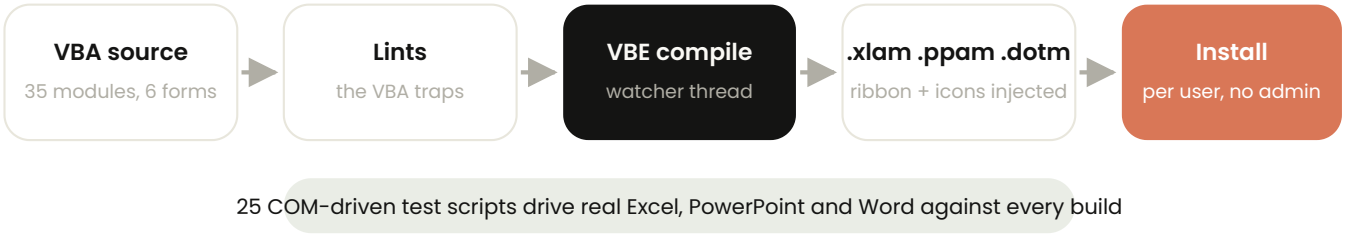


The slide. Align, stamp, check.

A build you can read, tests that drive real Office



The Excel tab. Workbook, Color, Model, Audit, Comps, Export, Lattice. Every large button carries one of the 40 generated icons.



Source

src\excel, **src\ppt**, **src\word** hold the modules and form code; **src\shared** is compiled into PowerPoint and Word. UserForms are built from code at build time, so the repository holds no binary form files. **ribbon*.xml** are the customUI definitions.

Build

build\build.ps1 drives Office through COM: imports the modules, builds the forms, compiles through the VBE with a watcher thread that reads and dismisses the compile dialog, saves the add-ins and injects the ribbon XML and icons with System.IO.Packaging. Static lints catch reserved identifiers, one-letter procedures, one-line If procedures and declarations after a procedure.

Tests

tests drives real instances with a dialog watcher: command sweeps, semantic checks of the formula tools, the stray-Excel-instance scenario, refresh and version prompts, the Trace window by mouse and keyboard, the case runner, chart annotations, stickers, and UI captures that never steal focus.

Install in a minute

1. Close Excel, PowerPoint and Word.
2. Run **build\install.ps1** (or **build\build.ps1** first to rebuild).
3. Open the apps: the Lattice tab is there. Per user, no admin rights: Excel's OPEN key, PowerPoint's own AddIns collection, Word's STARTUP folder.

Keys and settings are files

Excel shortcuts live in **%APPDATA%\Lattice\excel_shortcuts.txt** and regenerate the PDF manual. Settings are key/value pairs next to it, edited from the Settings dialog. Nothing phones home.

Alt K Excel Alt L PowerPoint, Word. Office keeps T, E, I, O, V and D for its 2003 menus, which is why the tab is not Alt+T.

NEXT

Workbook diff as a styled report, an Explorer pane for Word, and a native-chart route for Mekko and Gantt only if it can be done honestly.

REPOSITORY

On GitHub at github.com/AshWatter/Lattice. Source, build, tests, figures, this guide and the shortcut manual travel together, and all of them regenerate from scripts.

CREDITS

Specified and driven by Chiltern Capital Management; written with Claude; verified by Excel, PowerPoint and Word themselves.

160
EXCEL COMMANDS

227
POWERPOINT COMMANDS

40
GENERATED ICONS, ONE SCRIPT

FOUR LINTS THAT EACH SAVED A BUILD

Reserved identifiers: VBA is case-insensitive, so a variable called *nExt* or *lib* is a keyword. **One-letter procedures** collide with loop variables. **One-line If procedures** swallow their End Sub. **Declarations after a procedure** compile into "Variable not defined" three screens away from the cause. All four are caught by a regex before Office is even started.

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